



# Enea MANZI

## Curriculum Vitae

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### 🎓 | Education

- 09/2024 - 07/2026 ○ **Master's Degree in Cyber Security, University of Milan, (Milan, IT)**  
**Thesis:** "Design and Implementation of an API-Agnostic Tool for REST API Security Assurance"  
**Supervisors:** Marco Anisetti, Claudio Agostino Ardagna  
**Final Grade:** 110/110 cum laude
- 09/2021 - 10/2024 ○ **Bachelor's Degree in System and Network Security, University of Milan, (Milan, IT)**  
**Thesis:** "Design and Implementation of a Framework for Evaluating Distributed Systems"  
**Supervisors:** Marco Anisetti, Filippo Berto  
**Final Grade:** 110/110 cum laude
- 09/2016 - 06/2021 ○ **High School Diploma in Computer Science, IIS Badoni, (Lecco, IT)**  
**Final Grade:** 93/100

### 👜 | Work Experience

- 12/2025 - 07/2026 ○ **Curricular Internship for Master's Thesis, SESAR Lab, University of Milan, (Milan, IT)**  
*Secure Service-oriented Architectures Research Lab (SESAR Lab)*  
Designed and implemented **APIGuard Assurance**, an **API-agnostic**, **contract-driven** tool for automated REST API **security assurance** based on a taxonomy of eight domains and *specified oracles* to establish **trust** in automated verdicts, validated against a real-world target.
- 03/2024 - 09/2024 ○ **Curricular Internship for Bachelor's Thesis, SESAR Lab, University of Milan, (Milan, IT)**  
*Secure Service-oriented Architectures Research Lab (SESAR Lab)*  
Contributed to the design and implementation of an **Assurance Engine** for the verification of **non-functional properties (NFP)** in distributed systems, built in **Rust** and deployed on **Kubernetes**, developing a new connector to integrate **Elasticsearch** as an **observability** backend.
- 01/2020 - 02/2020 ○ **WBL: web developer, DaTriK Solutions, (Casargo, IT)**  
*High school work-based-learning*  
Developed a web application using HTML, JavaScript, and jQuery as part of a high school work-based learning program (PCTO).

## | Research Activities

12/2025 - 07/2026 ○

### SESAR Lab, University of Milan, (Milan, IT)

*Secure Service-oriented Architectures Research Lab (SESAR Lab)*

- Analyzed existing approaches to REST API testing, observing that most focus on **functional** and **syntactic** conformance, while **security violations** remain comparatively unaddressed.
- Defined a taxonomy of **eight security domains** to structure systematic assessment coverage.
- Proposed an **API-agnostic, contract-driven** methodology, based on *specified oracles*, that derives target knowledge exclusively from the **OpenAPI Specification (OAS)** and a deployment configuration file.
- Addressed the *oracle problem* in automated security testing, central to establishing **trust** in an automated verdict, by deriving, for each security control, an evaluation criterion from the specific knowledge and expectations of the test being performed, traceable to **OWASP** categories and **CWE** codes.
- Achieved **deterministic, reproducible** results, a prerequisite for assurance integrated into continuous development, through a **DAG**-based scheduling approach exposing **semantic exit codes** for **CI/CD** integration.
- Validated the methodology against a real-world target, **Forgejo** exposed through the **Kong** API gateway, confirming correct detection of security violations and deterministic, reproducible results.

03/2024 - 09/2024 ○

### SESAR Lab, University of Milan, (Milan, IT)

*Secure Service-oriented Architectures Research Lab (SESAR Lab)*

- Investigated existing approaches to verifying and monitoring **non-functional properties (NFP)** in distributed **Edge-Cloud Continuum** systems.
- Contributed to the development of an **Assurance Engine** for distributed systems, building a new connector to integrate **Elasticsearch** as an **observability** backend.
- Formalized contracts to jointly evaluate multiple non-functional properties (**CIA triad**, reliability, scalability, performance), validated experimentally on a **Kubernetes** testbed with minimal computational overhead.

## | Projects

### **APIGuard** *Master's Thesis Project (A.Y. 2025/2026)*

**Assurance** Development of an API-agnostic, contract-driven tool for REST API security assurance, with a particular focus on semantic security violations that traditional functional and syntactic testing does not capture.

- Built in **Python 3.12**, with a **dynamic test-discovery mechanism** allowing new security checks to be added without modifying the core engine.
- Implemented *specified oracles* directly in each test's code: every check encodes its own evaluation criterion, derived from the domain knowledge of the control under test.
- Enforced **strict static typing** (mypy) and **runtime validation** (Pydantic), with full documentation coverage across the public API.
- Built **reproducible packages** with **Hatch**, verifying identical SHA-256 hashes across repeated builds.
- Designed an **extensible** connector architecture to integrate external security scanners, decoupling test logic from tool-specific output parsing.

- Event-Driven IoT Architecture (Kafka + Kong)** *Cloud Computing Technologies Course Project (A.Y. 2025/2026)*  
 Development of a cloud-native microservices infrastructure on Kubernetes for IoT ingestion and monitoring, with a particular focus on **non-functional properties** such as scalability, fault tolerance, and edge-to-core security.
- Microservices architecture based on **Apache Kafka** in KRaft mode (ZooKeeper-less) for asynchronous decoupling and reliable message handling (*Zero Data Loss*).
  - Integration of **Kong API Gateway** to implement the *Gateway Offloading* pattern, centralizing Authentication (API Key), Rate Limiting, and Load Balancing at the edge.
  - Storage layer optimized on **MongoDB** using **Time Series Collections** and hybrid compression (LZ4/Zstd) to maximize storage efficiency and analytical performance.
  - Full **Kubernetes** orchestration managed via **Helm**, featuring automatic scalability (HPA) and advanced security (Secrets, TLS/SASL) to meet non-functional requirements.
- FTP Client and Server** *Computer Networks Course Project (A.Y. 2022-2023)*  
 Developed a multi-threaded **FTP-compliant** client-server application in **C** for file transfer, fully interoperable with **FileZilla**.
- Architecture: Multi-threaded server based on **Socket Programming** for concurrent session management.
  - Data Modes: Native support for both Active (PORT) and Passive (PASV) data transfer modes.
  - Protocol Management: Implemented essential *FTP commands* (e.g., USER, PASS, LIST, RETR, STOR, CWD...)
  - **Key skills: C, socket programming, multithreading, network protocols, FTP, Git.**
- Exhibition Web App** *Web and Mobile Programming Course Project (A.Y. 2021-2022)*  
 Built a dynamic web application serving as a digital portfolio for displaying multimedia content.
- Developed the backend with *Node.js/Express* and the frontend with *HTML5, CSS3, JavaScript/jQuery*.
  - Integrated *MongoDB* for document-oriented storage and *AJAX* for asynchronous client-server communication (GET, POST, DELETE).
  - Managed source code and collaboration using **Git** and **GitHub**.
  - **Key skills: Node.js, Express, MongoDB, AJAX, REST APIs, jQuery, Git.**

## Publications

- Jun 2026 ○ **Security Assurance for Modern Big Data APIs**  
*Enea Manzi, Marco Anisetti*  
*[Submitted to ITADATA 2026]*  
 Automated REST API testing has traditionally focused on functional compliance, leaving security assurance comparatively underexplored; existing tools operate at the syntactic level and cannot reliably detect the semantic vulnerabilities behind several top-ranked OWASP API Security categories. This work proposes a contract-driven approach that derives all target knowledge from the OpenAPI specification and a configuration file, embedding evaluation criteria directly into deterministic test oracles and organizing checks into an eight-domain taxonomy scaling with available privileges. Tested on a realistic setup, the approach correctly classified expected outcomes, revealed a real discrepancy between declared and actual API behavior, and produced reproducible results, contributing a systematic approach to the oracle problem and a process ready for integration into automated pipelines.

## Languages

Italian Mother tongue

English Intermediate *Listening B2 - Reading B2 - Writing B2 - Spoken Production B2 - Spoken Interaction B2*

## | Skills

|                          |                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages    | C, Java, Python, SQL, Web (HTML, JavaScript/Node.js), OOP Concepts, low-level programming                                           |
| Database                 | MySQL, Database Design, Implementation and Administration                                                                           |
| Architectures & Software | Docker, Kubernetes, microservices architectures, REST API, Git, Kafka, Kong API Gateway, CI/CD, Observability, Edge-Cloud Continuum |
| Operating Systems        | Fresh OS installation, Linux Administration and Shell, Windows                                                                      |
| Networking & Security    | Network Protocols, ICT Network Security Risk, Secure Coding Practices, API Security, Continuous Security Assurance                  |
| Tools                    | Visual Studio Code, GitHub                                                                                                          |

## | Soft Skills

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Thinking      | analytically, critically, proactively                        |
| Problems      | address problems critically, solve problems/create solutions |
| Communication | cooperate with colleagues, work in teams, manage teamwork    |
| Others        | demonstrate curiosity, cope with stress, prioritise tasks    |

*Bellano, July 21, 2026*

*Enea Manzi*